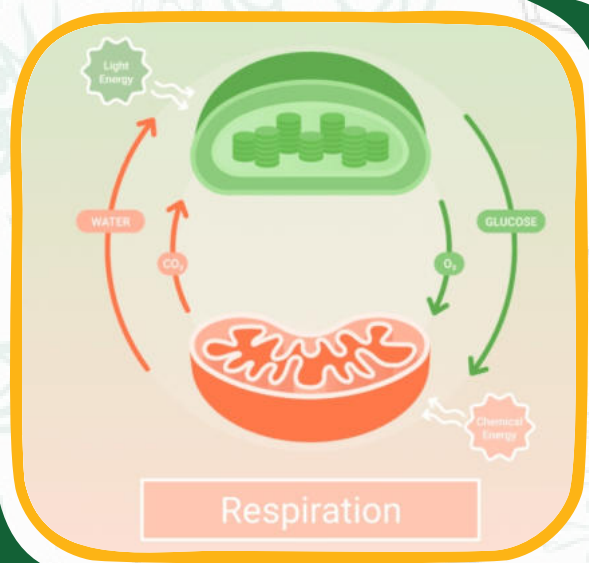
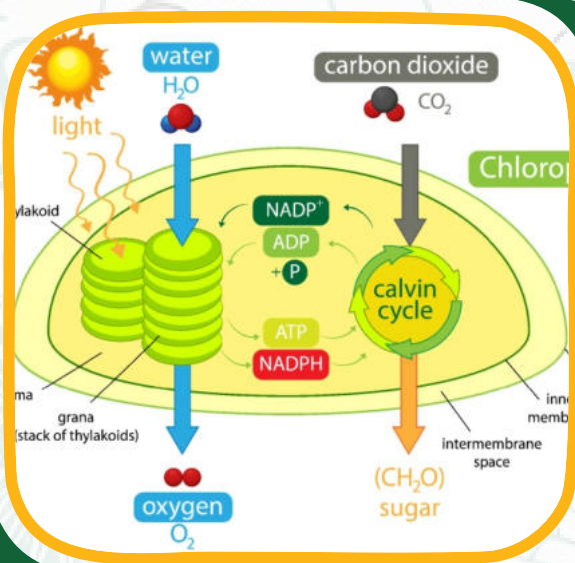




BIOENERGETICS



Important Definitions:**1. Bioenergetics:**

The study of conversion of free energy into different form by a living organism

2. Stroma:

Chloroplast have semi-fluid proteins containing medium is called stroma.

3. Photosystem:

A biochemical mechanism in which plants absorb light energy for photosynthesis.

3. Redox Reaction:

The oxidation and reduction occurring together is called redox reaction.

4. Photolysis:

The breakdown of any particular substance in the presence of sunlight.

5. Granum:

Piled up thylakoid in a form of stalk is called grana.

Energy**Definition:**

Energy is the ability to do work.

Energy Conservation:**Definition:**

A change from one form of energy into another is called energy conservation.

Energy in Living Organism:

A living cell exhibits ceaseless chemical activities. Substances are broken down inside the cells and new substances are formed. Living organisms are also like machines which require nutrients in the form of food. The special molecules of food contain energy. This energy comes in fuel and food molecules. Energy consists in two forms in living organism.

- kinetic Energy
- Potential Energy

I. Kinetic Energy**Definition:**

The energy of a body due to its motion is called kinetic energy.

II. Potential Energy:

The energy stored in an object due to its relative position or internal structure.

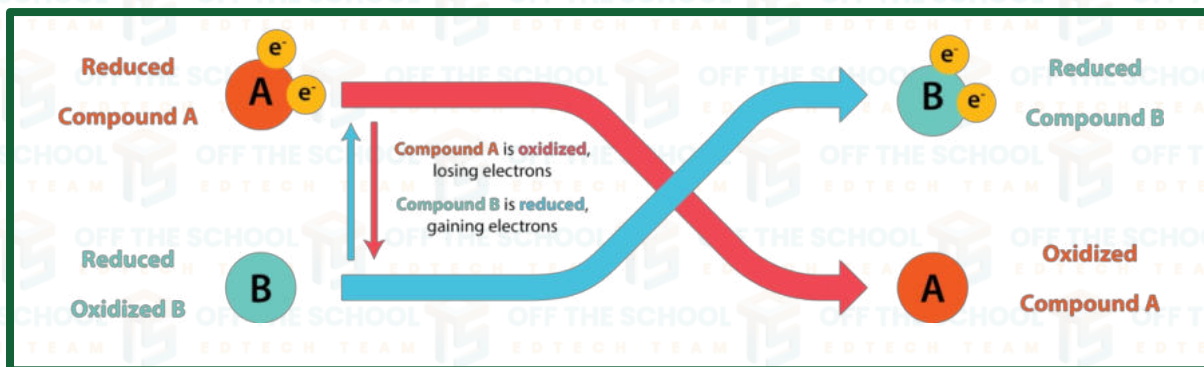
Bioenergetics**Definition:**

The study of conversion of free energy into different forms by living organisms is called Bioenergetics.

Transfer of Energy to Living Organisms:

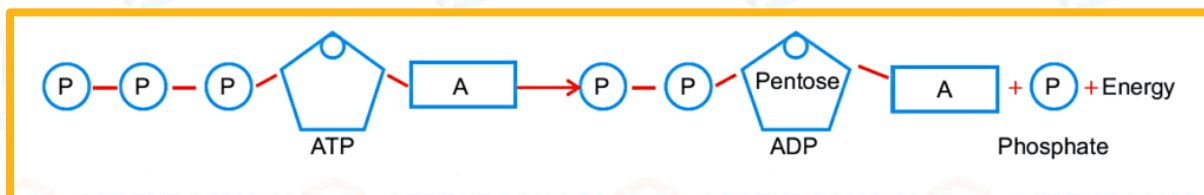
Organisms obtain energy by metabolizing the food they eat or prepare. Food contains potential energy in its bond. When these bonds break down, a large amount of kinetic energy is usually released. Some of this energy is stored in the form of potential energy in the bonds of ATP molecules while the rest escapes as heat. The potential energy stored in ATP is again transformed into kinetic energy to carry out life activities. In living organisms' energy is transferred through gain or loss of electrons during formation and breaking of chemical bonds. There are two chemical process where it occurs.

- Oxidation
- Reduction



Energy Currency in Living Organisms:

- Living organisms have special type to store energy.
- These energy storing molecules are called Adenosine Tri-Phosphate (ATP).
- The ability of ATP to store and release energy is due to its molecular structure.
- Energy released any redox reaction is utilized by molecules called Adenosine -Di-Phosphate (ADP) to form bond with phosphate (P).
- As a result, ADP become ATP and energy of oxidation is now stored in ATP.



Photosynthesis

Definition:

The process by which plants make their own organic food by taking inorganic molecules from their surroundings, in the presence of sunlight is called photosynthesis.

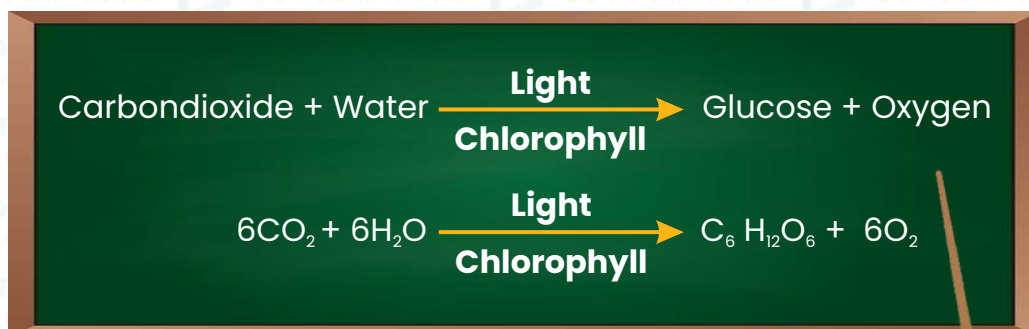
Formation of Glucose

Plants can build sugar by taking CO_2 from air and water from the soil, For the synthesis of sugar like glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) carbon and oxygen are provided by the atmosphere carbon dioxide whereas the water molecules provide hydrogen.

Procedure

Sunlight provides the energy which is stored in chlorophyll of the plant cells. Enzymes are also needed for this process. This process is completed in the mesophyll cells of the leaves where chlorophyll is present. In this the plants build sugar molecules from the simple inorganic compounds carbon, oxygen and hydrogen.

Chemical Equation:



Importance or significance of photosynthesis:

(i) Role of Glucose:

The fundamental molecule produced during photosynthesis is simple sugar i.e. Glucose. Glucose utilizes in most of the metabolism of plant to produce secondary products like starch and other polysaccharides. Plants also use carbohydrates to form fats, proteins and other chemical like Nucleic Acids. This glucose is also used in respiration as reactant to produce energy for the metabolism of living organisms.

(ii) Essential in Food Chain:

Plants are not the only organisms which depend on photosynthesis but animals (Heterotrophs) also depend on phototrophs (i.e. plants). Animals utilize the molecules of phototrophs as food molecules. If an animal is herbivorous it feeds directly on plants. If an animal is carnivorous it depends on herbivores (those animals which feed on plants). These feeding sequences and relationship are called Food Chains.

(iii) Role of Oxygen:

Photosynthesis is the only process which produces free O_2 by splitting water. This O_2 is utilized by all living organisms for respiration to produced energy for metabolism. Without O_2 living organisms cannot survive.

(iv) Quantity of CO_2 and O_2 in nature Is Maintained:

During Photosynthesis plants fix CO_2 and release O_2 in environment. CO_2 has a property to absorb heat of sun. If its quantity increases in environment, there will be increase in an environmental temperature on earth called Global warming. Thus, photosynthesis keeps the quantity of CO_2 to maintain the temperature of earth

Role of Chlorophyll in Photosynthesis:

Chlorophyll is the green pigment found in the chloroplast of plant cells. It captures a specific part of visible light only, therefore it is not a reactant but absorbs energy needed to drive the reaction. In other words, Photosynthesis converts light energy to chemical energy.

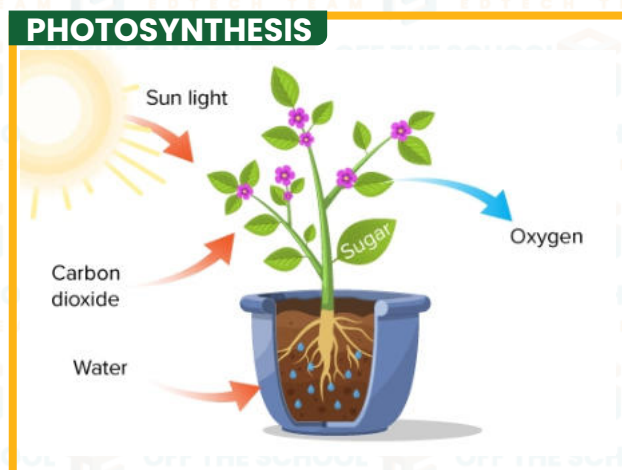
Chloroplast, a Light Trapping and Storage Organelle

Green parts of plants and algae contains special cell organelle called chloroplast.

Structure of Chloroplast:

- Chloroplast are oval shaped.
- Chloroplast has two membranes an outer and an inner.
- The space within the inner membrane is called the stroma.
- Chloroplast have many small disc-shaped sacs called thylakoids within their stroma.
- In vascular plants and green algae, the thylakoids are stacked on top of one another, and a stack of thylakoids is called a granum (plural: grana).
- The thylakoids contain chlorophylls and carotenoids, and these pigments absorb light during the process of photosynthesis.

Reaction of photosynthesis involves a number of chemical reactions which are catalyzed by a number of enzymes, either in cyclic or noncyclic ways. Each reaction occurs are different site in chloroplast limelight dependent reaction and dark independent reaction.



Phase of Photosynthesis

The whole process of photosynthesis is mainly divided into two stages

- Light Reaction OR Light –Dependent Reaction
- Dark Reaction OR Dark– Dependent Reaction

Light Reaction:

Light reaction is also known as light dependent reaction, as it requires light in the reaction. Chlorophyll and other molecules capture energy from light and convert some of it into chemical energy.

PHOTOLYSIS:

In the light reaction, light energy is trapped by chlorophyll. This light energy is used to split water molecules into oxygen and hydrogen. This splitting energy is known as **“photolysis”** (photo=light, lysis= to break)

RELEASE OF OXYGEN:

Oxygen which is produced during photolysis is released in the environment and hydrogen which is also produced during photolysis is used, together with CO_2 in the synthesis of glucose. Conversion of light energy into chemical energy:

Chlorophyll and other pigments molecules within the chloroplast, form highly-organized assemblies known as “photo systems”. Conversion of light energy into chemical energy in these photosystems produces two energy rich compounds which are:

- a) NADPH₂ Nicotinamide Adenine Dinucleotide Phosphate
- b) ATP (Adenosine Triphosphate)

Formation of NADPH₂:

NADP already exists in the chloroplast reduced into NADPH₂.

Equation:



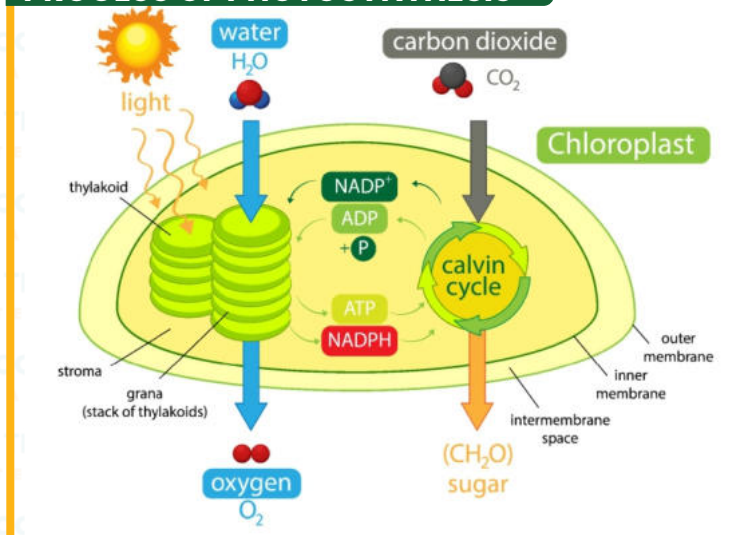
Formation of Adp:

ADP is another compound which is present in the cell. ADP (Adenosine Triphosphate) reacts with the phosphate group by using the light energy to form ATP.

Equation:



PROCESS OF PHOTOSYNTHESIS



Dark Reaction:

Dark reaction is also known as the light independent reaction because light energy is not captured during this phase of photosynthesis.

Mechanism of Dark Reaction:

NADPH₂ and ATP are the two energy rich compounds which are formed during the light reaction. These compounds provide energy for the combination of CO₂ and H₂O to synthesize glucose. Fixation of CO₂ and its conversion into glucose occurs in the chloroplast by means of a series of reactions known as Calvin cycle of dark reactions. In this stage of photosynthesis glucose molecules are formed and stored in the chloroplast such as starch.

Limiting Factors:

Definition:

Any environment factor's absence or deficiency of which can decrease the rate of a metabolic reaction is called the limiting factor.

Many factors act as limiting factors for photosynthesis.

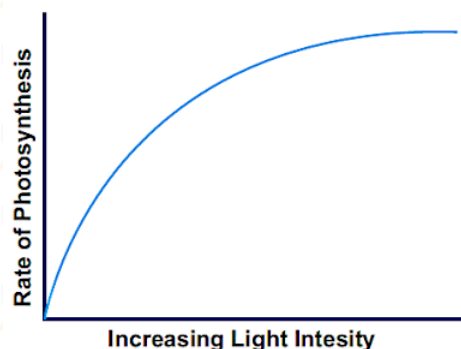
i. Light Intensity

ii. Temperature

iii. Carbon Dioxide

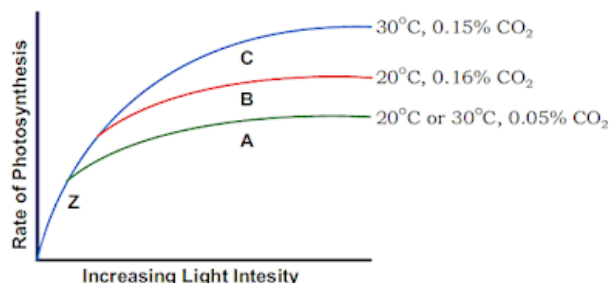
i) Light Intensity:

The rate of photosynthesis varies with light intensity. It decreases as the light intensity and increases as the intensity increases. However, at much higher light intensity, the rate of photosynthesis becomes constant.



ii) Temperature:

The rate of photosynthesis decreases with decreases in temperature. It increases as the temperature is increased over a limited range. But if light intensity is low, increasing the temperature has little influence on the rate of photosynthesis.



iii) Carbon Dioxide:

Carbon dioxide concentration raises the rate of photosynthesis, it goes on increasing until limited by the factors. Increases in CO₂ concentration beyond a certain level causes the closure of stomata and it decreases the rate of photosynthesis.

Respiration

Definition:

A chemical process in which oxygen is used to make energy from carbohydrates (sugars).

Chemical Equation:

Glucose + Oxygen → Carbon dioxide + water + Energy



Cellular Respiration:

Definition:

It is the oxidation of food taking place within the cells by the help of oxidative enzymes resulting in the release of energy. Cellular respiration also called tissue respiration takes place within the cell. It is the actual process which liberates energy.

Types of Respiration:

There are two types of respiration found in living organisms for the production of energy.

i. Aerobic Respiration

ii. Anaerobic Respiration

Anaerobic Respiration:

Definition:

Anaerobic respiration also called "Fermentation". It takes place in the absence of oxygen; hence the food is oxidized partially, releasing a small amount of energy. Only (5% to 10%) as compared to aerobic respiration. It takes place in some bacteria, fungi, yeast and endoparasites. In anaerobic respiration, glucose is not broken down completely so less energy is released but it can sustain life even in the absence of oxygen.

There are two types of anaerobic respiration

i) Alcoholic Fermentation ii) Acidic Fermentation

Chemical Equation:

Glucose → Ethanol + CO₂ + Some energy



Acidic Fermentation:

Acidic Fermentation is a type of cellular respiration which does not require oxygen (**anaerobic respiration**), and involves the breakdown of glucose to lactic acid. During glucose breaks down into a substance called lactic acid. In acidic Fermentation a limited amount of energy is produced as compared to aerobic respiration.

Chemical Equation:

Glucose → Lactic acid + Some energy



Importance of Anaerobic Respiration (Fermentation):

The following points explain the economic importance of the process of fermentation:

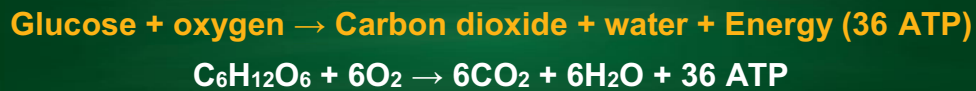
1. It is a source of various products such as ethyl alcohol, lactic acid, propionic acid and butanol etc.
2. The process is used in brewing industries for the production of wine.
3. Dairy industries use this process for the production of various dairy products such as cheese, yoghurt etc.
4. In baking process, the yeast fermentation is utilized
5. Several industrially important solvents such as acetone are the by-products of fermentation.
6. Lactic acid and acetic are the products of Fermentation which are used in various fields.
7. The soya sauce is made through fermentation by a fungus *Aspergillus*.
8. Humans can also provide energy to their skeletal muscle cells through anaerobic respiration.

2. Aerobic Respiration:

Definition:

It takes place in the presence of free oxygen, oxidizing the food completely and releasing the maximum energy, 2827 kJ/mole of glucose. The end products are carbon dioxide and water.

Chemical Equation:



Mechanism of Aerobic Respiration:

Aerobic respiration takes place in three steps at different places in the cell

1. Glycolysis
2. Krebs's Cycle
3. Electron Transport Chain (ETC)

a) Glycolysis:

Gr: Glycol = Sugar
Lysis = Break down

"The process in which a glucose molecule (six carbon sugar) is broken down into two molecules of Pyruvic acid (3C acid)".

Glycolysis occurs in cytoplasm and oxygen is not involved at this stage, so it occurs in both

aerobic and anaerobic respiration. This splitting of glucose releases a small amount of energy of glucose which is enough to generate 2 molecules of ATP. Glycolysis is a complex sequence of reactions all occurring in cytosol.

b) Krebs's Cycle:

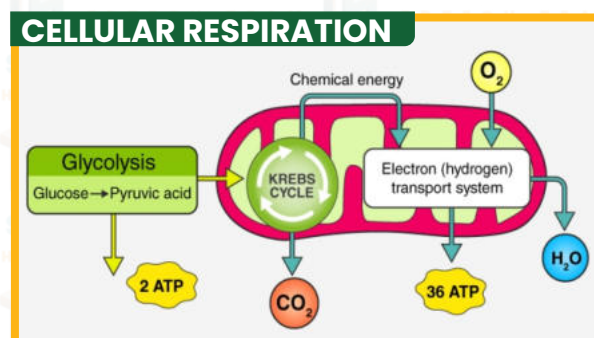
A British biochemist, Sir Hans Krebs discovered this series of reactions that is why it is called the Krebs cycle. In the Krebs cycle, pyruvic acid molecules are completely oxidized along with the formation of ATP, NADH and FADH₂. Before entering in Krebs cycle, pyruvic acid is changed into 2-carbo compound called acetyl CoA.

c) Electron Transport Chain (ECT):

It is the last stage of cellular respiration. It is the transfer of electrons in an electron transport chain. In this step

- NADH and FADH₂ release electrons and hydrogen ions.
- These electrons are taken up by a series of electron carriers.
- When electrons move, through the series of electron carriers they lose energy, which is used to synthesize ATP.
- At the end of the chain, electrons and hydrogen ions combine with molecular oxygen and form water compound called acetyl CoA.

Use of Respiration Energy in Body of an Organism:



- Plants and animals need energy to stay alive. Following are some processes which utilize respiration energy.
- Energy is used by all organism to build up large molecules from smaller ones (for example, building up proteins from amino acids or making starch from glucose needs energy.)
- Energy is used by animals to contract muscles during movement.
- Energy is used by mammals and birds to keep their body temperature steady in colder surroundings.
- Organisms use energy for their growth and repair of cells.
- Respiratory energy is used in movement of ions and molecules from low concentration to high.
- During cell division formation of large molecules (DNA and protein) and movement of chromosomes require energy.
- Energy is used in conduction of nerve impulses (Electrical signals of neurons) which takes place in the active transport of ions.

Difference Between Photosynthesis and Respiration

S.no	PHOTOSYNTHESIS	RESPIRATION
01	Photosynthesis is the process where light energy is converted into chemical energy	Respiration is the process where chemical energy converted into energy of ATP.
02	It occurs only in chlorophyll containing organisms.	It occurs in all organisms.
03	It requires light sources i.e. occur only in the presence of light.	It does not require a light source so it occurs throughout life.
04	It occurs in the chloroplast.	It occurs in the mitochondria
05	The reactants are carbon dioxide and water.	Reactants are carbohydrates and oxygen usually.
06	Products are glucose and oxygen.	Products are carbon dioxide and water in the case of aerobic respiration.

Difference Between Aerobic Respiration and Anaerobic Respiration:

S.no	PHOTOSYNTHESIS	RESPIRATION
01	Oxygen is necessary for the oxidation.	Enzymes are used for oxidation.
02	Food is completely oxidized.	Food is partially oxidized.
03	Metabolic wastes are products.	Toxic substances like alcohol or lactic acid are produced.
04	Maximum quantity of energy is released.	Lesser quantity of energy is released.
05	It is found in most organisms.	It is found mostly in bacteria and fungi and endoparasites.